
Spectral Grouping and Drift Correction for Diffusion Model Quantization

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Abstract

Diffusion models achieve strong image generation performance by iteratively denoising a sample until it conforms to the target data distribution. Despite their effectiveness, sampling remains computationally expensive because architectures such as DDIM require repeated evaluations of a U-Net noise predictor over many time-steps. This cost limits deployment in edge and resource-constrained environments, motivating the use of model quantization for compression and inference acceleration. However, post-training quantization of diffusion models is challenging due to timestep-dependent activation distributions and the accumulation of quantization error across the denoising trajectory, both of which can substantially degrade generation quality. We introduce a spectral grouping method for post-training quantization of diffusion models on CIFAR-10. Rather than assigning quantization scales uniformly across time-steps, our approach groups time-steps according to spectral properties and allocates scales in a manner better aligned with the dynamics of the denoising process. In addition, we conduct two ablation studies: incorporating a drift correction term into the DDIM update equation to reduce error accumulation, and learning time-conditioned rotations to suppress activation outliers. Our results demonstrate that timestep-aware spectral grouping can improve the robustness of low-bit diffusion quantization while preserving image quality.

1. Introduction

Diffusion models have become one of the dominant paradigms for image generation due to their ability to synthesize high-quality and diverse samples through an iterative denoising process. Starting from a noisy input, these models gradually remove noise over a sequence of time-steps until the sample approaches the distribution of real images. In commonly used diffusion frameworks such as DDPM and DDIM, this reverse process is implemented using a U-Net backbone that predicts the noise component at each timestep.

The predicted noise is then used to update the current noisy sample and move it closer to a clean image. While this iterative formulation is highly effective for generation quality, it is computationally expensive because the U-Net must be evaluated repeatedly across tens, hundreds, or even thousands of denoising steps. This makes diffusion sampling costly in terms of latency, memory, and compute, limiting deployment in edge or resource-constrained settings.

Post-training quantization (PTQ) is a natural approach for reducing the deployment cost of diffusion models. By representing weights and activations with low-bit integers instead of full-precision floating-point values, PTQ can reduce memory usage and accelerate inference without requiring expensive end-to-end retraining. However, diffusion models present challenges that are not typically encountered in standard classification or language-model quantization. In particular, the activation distributions inside the U-Net are highly timestep-dependent. Early in the reverse process, the model operates on inputs that are close to Gaussian noise, while later time-steps contain increasingly structured image content such as edges, textures, colors, and object-level information. As a result, activation statistics collected at one region of the denoising trajectory may be poorly matched to those observed elsewhere. A single static quantization scale can therefore be too small for some time-steps, causing clipping, or too large for others, causing excessive rounding error.

A second major difficulty is that quantization error accumulates across the denoising trajectory. Unlike feed-forward image classification, where quantization error affects only one forward pass, diffusion sampling repeatedly feeds the output of one denoising step into the next. Consequently, even small quantization errors introduced at an early timestep can shift the sample away from the full-precision trajectory and influence all subsequent denoising updates. This creates a compounding effect: the quantized model is calibrated on full-precision or clean intermediate states, but at inference time it receives its own quantized and potentially drifted outputs. Prior work has identified this as both an error propagation problem and an exposure bias problem, making final image quality a more important measure than isolated per-step reconstruction error.

Several recent methods have attempted to address these challenges. Q-Diffusion introduced one of the founda-

tional PTQ pipelines for diffusion U-Nets by performing timestep-aware calibration across the full denoising trajectory and proposing split shortcut quantization to handle the mismatched distributions that arise in U-Net skip connections. This showed that calibration must account for the full sampling trajectory rather than a small set of randomly selected activations. However, Q-Diffusion still relies on static quantization scales after calibration, meaning that the final scale does not dynamically adapt to the timestep or the sample-specific activation distribution encountered during inference. It also reduces per-step error indirectly but does not explicitly model how quantization error propagates across time-steps.

Other methods attempt to improve the choice of calibration time-steps or quantization intervals. PTQ4DM proposes Normally Distributed Timestep Calibration, which oversamples regions of the trajectory where activation distributions are more extreme. APQ-DM further improves timestep selection by grouping time-steps and using structural risk minimization to select informative calibration points. These methods recognize that different parts of the trajectory require different quantization behavior, but their grouping strategies are still largely based on timestep indices, calibration heuristics, or activation statistics. Even when time-steps are grouped, the resulting quantization scales remain static or piecewise static within each group. Thus, the core issue of continuously varying activation distributions is only partially addressed.

Temporal Dynamic Quantization (TDQ) takes a different direction by learning a timestep-conditioned function for predicting quantization intervals. Instead of using one fixed scale, TDQ trains a lightweight MLP that maps the timestep to a quantization interval, often using Fourier features to represent time. This demonstrates that continuous timestep-aware scaling is feasible and can be precomputed to avoid runtime overhead. However, TDQ conditions only on the scalar timestep. Two samples at the same timestep receive the same scale even if their activations have different magnitudes, outliers, or channel-wise statistics. In diffusion models, where both timestep and sample content influence the U-Net’s internal activations, conditioning only on time leaves a significant gap.

Outlier suppression is another important direction for quantization. SmoothQuant shows that activation outliers can make joint weight-activation quantization difficult, and proposes shifting part of the quantization difficulty from activations to weights through a mathematically equivalent per-channel smoothing transformation. Although originally proposed outside diffusion models, this idea is relevant because diffusion U-Nets also suffer from activation outliers that can dominate scale selection. However, SmoothQuant computes smoothing factors from a static calibration set and assumes

that outlier structure is relatively stable. This assumption is weak in diffusion models, where activation distributions and outlier patterns change across time-steps. Similarly, rotation-based methods such as SpinQuant suggest that learned rotations can suppress outliers by redistributing activation energy across dimensions, but existing rotation approaches are not designed to adapt with respect to diffusion timestep. This leaves open the question of whether time-conditioned rotations can better suppress timestep-dependent outliers in diffusion U-Nets.

Despite this progress, several gaps remain. First, existing timestep grouping methods do not exploit the spectral structure of the denoising process. The reverse diffusion trajectory naturally changes the frequency content of the sample: early noisy states are dominated by different spectral characteristics than later, more structured image states. Yet prior PTQ methods do not group time-steps according to the frequency content of activations or intermediate noisy samples. Existing grouping is therefore not directly tied to the representational structure of the denoising trajectory. Second, current dynamic scaling methods are either static within groups or conditioned only on timestep. They do not jointly condition quantization scales on both time and the actual activation statistics observed at that step. Third, although outlier suppression has been studied through smoothing and rotations, existing methods do not learn rotations that vary with diffusion time, even though outlier behavior itself changes across the denoising process. Finally, quantization error accumulation remains a persistent issue: calibration reduces local error, but the drift between full-precision and quantized trajectories can still compound over many denoising steps.

In this work, we propose a post-training quantization framework for diffusion models that directly targets these gaps. Our central idea is to group time-steps using spectral information rather than relying only on timestep index or empirical activation clustering. Specifically, we use frequency-domain characteristics of the denoising trajectory to form groups that better reflect the changing structure of the sample and the corresponding activation regimes inside the U-Net. Within each spectral group, we allocate quantization scales in an activation-aware manner so that the scale is better matched to the statistics of the current timestep and sample. This allows the quantizer to respond not only to where the model is in the denoising trajectory, but also to what distribution the U-Net is actually processing.

We further study two complementary mechanisms. First, we introduce a drift correction term in the DDIM update equation to reduce the accumulation of quantization error across time-steps. Rather than assuming that minimizing local reconstruction error is sufficient, this correction explicitly targets the trajectory mismatch between the full-

precision and quantized sampling processes. Second, we investigate learned time-conditioned rotations for outlier suppression. Inspired by the motivation behind SmoothQuant and rotation-based quantization, these rotations aim to reduce timestep-dependent activation outliers before quantization, while allowing the transformation to vary across the denoising trajectory.

The goals of this paper are therefore as follows:

- To develop a spectral grouping strategy for diffusion PTQ that partitions time-steps according to frequency-domain structure rather than only timestep index or activation-statistic heuristics.
- To allocate quantization scales in a timestep- and activation-aware manner so that quantization better adapts to the non-stationary activation distributions of diffusion U-Nets.
- To reduce quantization-induced trajectory drift by incorporating a drift correction term into the DDIM sampling update.
- To study time-conditioned rotations for outlier suppression and evaluate whether timestep-dependent transformations improve low-bit quantization robustness.
- To evaluate the proposed method under a post-training setting on CIFAR-10, comparing against core baselines such as Q-Diffusion and a few ablation studies for our drift correction approach and application of time dependent rotations.

2. Methodology

Diffusion models are distinct from other image-based models, such as Vision Transformers, in the sense that they use a special, multi-timestep approach at inference. In this section, we discuss the challenges that this architecture presents, and we elaborate on our novel quantization scheme.

2.1. The Challenges of Denoising Models

2.1.1. VARYING ACTIVATION DISTRIBUTIONS

The multi-timestep approach of denoising models makes them especially difficult to quantize to lower bit sizes, since the layer activations vary significantly throughout the denoising process. Thus it is difficult to produce a scale s_k and a zero point z_k for each layer k which accurately represents the activation distribution that the layer will produce throughout the entire denoising trajectory.

As seen in Fig. 1, the vast majority of the images produced in the trajectory are highly noisy, with the low variance features coming in rapidly towards the end. This means

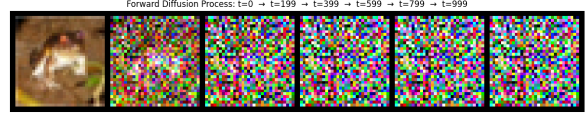


Figure 1. Regular Forward Denoising Trajectory

our quantization scale will be highly skewed towards the earlier, noisy images. Papers like PTQ4DM attempt to solve this issue by sampling time-steps from a normal distribution skewed towards the lower time-steps, and generating a calibration set from these sampled time-steps. Although this is effective, and produces far better results than the randomly sampled calibration set, it is not entirely "timestep-aware". The information is encoded within the scales, but they still apply a single uniform scale to every time-step.

Moreover, these varying activation distributions exacerbate the issue of outliers; with N time-steps, there are N possible activation distributions, which can all produce a significant number of outliers. Quantizing a diffusion model thus requires extra care in the suppression of outliers.

2.1.2. ACCUMULATION OF ERROR ACROSS TIME-STEPS

Quantizing the weights and activations of a diffusion model requires a rounding operation, which injects a certain amount of quantization noise within the full-precision weights and activations. Since the output x_{t+1} at each time-step t is entirely conditioned on the output of the previous layer x_t , the quantization noise accumulates through both layers.

Suppose a function f_θ predicts the next state x_{t+1} given the current state x_t and timestep t :

$$x_{t+1} = f_\theta(x_t, t).$$

Let $\hat{f}_\theta$ denote the quantized version of the same transition function. The quantized trajectory is therefore

$$\hat{x}_{t+1} = \hat{f}_\theta(\hat{x}_t, t).$$

We define the accumulated trajectory error at timestep $t + 1$ as

$$e_{t+1} = \hat{x}_{t+1} - x_{t+1}.$$

Substituting the full-precision and quantized transition functions into this definition gives

$$e_{t+1} = \hat{f}_\theta(\hat{x}_t, t) - f_\theta(x_t, t).$$

To separate the error caused directly by quantization from the error caused by propagating an already perturbed input, we add and subtract $f_\theta(\hat{x}_t, t)$:

$$e_{t+1} = [\hat{f}_\theta(\hat{x}_t, t) - f_\theta(\hat{x}_t, t)] + [f_\theta(\hat{x}_t, t) - f_\theta(x_t, t)].$$

The first term is the local quantization error at timestep t :

$$\epsilon_q^{(t)} = \hat{f}_\theta(\hat{x}_t, t) - f_\theta(\hat{x}_t, t).$$

Assuming the trajectory error

$$e_t = \hat{x}_t - x_t$$

is small, the second term can be approximated by a first-order Taylor expansion around the ideal input x_t :

$$f_\theta(\hat{x}_t, t) - f_\theta(x_t, t) \approx J_t(\hat{x}_t - x_t) = J_t e_t,$$

where

$$J_t = \nabla_x f_\theta(x_t, t)$$

is the Jacobian of the denoising transition with respect to the input at timestep t . Substituting this approximation gives the recurrence

$$e_{t-1} \approx J_t e_t + \epsilon_q^{(t)}.$$

Assuming the initial noise is the same for both models, so that

$$e_T = 0,$$

the total error at the end of generation can be written as

$$e_0 \approx \sum_{t=1}^T \left(\prod_{k=1}^{t-1} J_k \right) \epsilon_q^{(t)}.$$

Thus, we observe two important properties:

1. The quantization noise $\epsilon_q^{(t)}$ injected at each timestep contributes to the final trajectory error.
2. Noise from earlier denoising steps can be repeatedly transformed and amplified by the Jacobians of subsequent steps. If the dominant eigenvalues of these Jacobians are greater than one, small local quantization errors may lead to large deviations in the final generated sample.

2.2. Post Training Quantization on Denoising Model

We primarily propose an inherently "timestep-aware" method of quantization through sampling via spectral grouping. Moreover, to solve the issues of quantization error accumulation and outliers, we propose a quantization drift predictor and a variant of SpinQuant respectively.

Spectral Grouping. There is a growing body of research dedicated to the "spectral bias" and "frequency evolution" of diffusion models, showing that diffusion models operate, on some level, in the frequency (Fourier) domain. Thus, we are able to extrapolate information about the distributions of the activations based on the frequency content observed over a series of samples. We propose that spectral content is

an effective measure of similarity between time-steps, and that spectral content alone can effectively be used to create quantization "bins" to dedicate a common scale.

We empirically measure the shift from low-frequency to high-frequency generation by calculating the Spectral Centroid of the network's intermediate representations.

Let $A^{(t)} \in R^{B \times C \times H \times W}$ denote the feature map activations extracted from the network at diffusion timestep $t \in [0, T]$, where T is the initial state of pure noise and 0 is the fully denoised sample. For a specific batch index b and channel c , the 2D Power Spectral Density (PSD) is computed via the discrete Fourier transform (DFT):

$$S_{b,c}^{(t)}(u, v) = \left| \sum_{x=0}^{W-1} \sum_{y=0}^{H-1} A_{b,c}^{(t)}(x, y) e^{-j2\pi \left(\frac{ux}{W} + \frac{vy}{H} \right)} \right|^2$$

where u and v represent the spatial frequencies along the width and height dimensions, respectively. The zero-frequency (DC) component is shifted to the geometric center of the frequency grid. The expected 2D PSD is then obtained by averaging across the batch and channel dimensions:

$$\bar{S}^{(t)}(u, v) = \frac{1}{B \cdot C} \sum_{b=1}^B \sum_{c=1}^C S_{b,c}^{(t)}(u, v)$$

Because natural images exhibit uniform frequency characteristics, we reduce the 2D PSD into a 1D radial power profile. We define the discrete radial frequency distance as $r = \lfloor \sqrt{u^2 + v^2} \rfloor$. The average spectral power at a given radial distance r is formulated as:

$$P^{(t)}(r) = \frac{1}{|M_r|} \sum_{(u,v) \in M_r} \bar{S}^{(t)}(u, v)$$

where $M_r = \{(u, v) : \lfloor \sqrt{u^2 + v^2} \rfloor = r\}$ defines the set of Cartesian frequency coordinates that map to the radius r , and $|M_r|$ is the cardinality of that set.

To summarize the frequency distribution into a single comparative scalar, we calculate the Spectral Centroid $C(t)$. To prevent the average image intensity from skewing the distribution, the direct current (DC) component ($r = 0$) is strictly excluded. The centroid is computed as the power-weighted mean of the spatial frequencies:

$$C(t) = \frac{\sum_{r=1}^{r_{max}} r \cdot P^{(t)}(r)}{\sum_{r=1}^{r_{max}} P^{(t)}(r) + \epsilon}$$

where $r_{max} = \min(H/2, W/2)$ corresponds to the spatial Nyquist limit, and ϵ is a small scalar added for numerical stability.

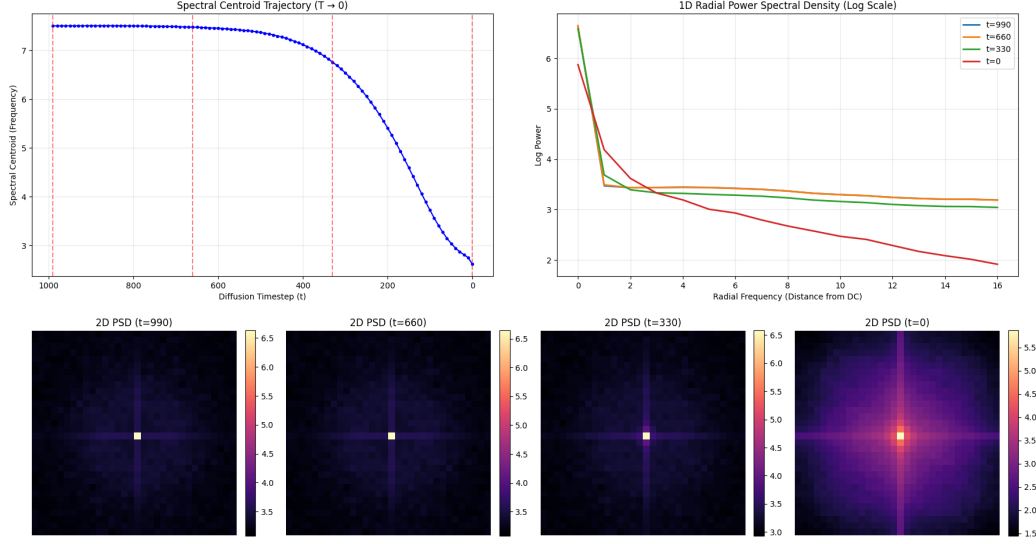


Figure 2. Spectral Data

Fig.2 illustrates the empirical spectral data of the diffusion model’s spatial activations, confirming a pronounced autoregressive spectral bias across the reverse trajectory ($T \rightarrow 0$). At early time-steps ($t \approx 1000$), the high spectral centroid, flat 1D radial power profile, and diffuse 2D PSD heatmaps collectively indicate a noise distribution heavily weighted by high frequencies. As the denoising process unfolds, the spectral centroid exhibits a rapid, monotonic decay, while the 1D PSD transitions to a steep $1/f$ -like decay characteristic of natural images, and the 2D spectral energy tightly clusters around the low-frequency center. Due to the monotonic nature of the centroid frequency, we can easily group different time-steps based on their proximity in centroid frequency.

To do so, we aim to group the time-steps by centroid frequency, while also preventing overfitting in the form of a high number of groups. To address this, we partition the denoising trajectory into discrete groups using optimal change-point detection.

Let the ordered sequence of spectral centroids evaluated over N reverse diffusion time-steps be denoted as a 1D time-series signal $Y = \{y_1, y_2, \dots, y_N\}$. Our objective is to partition Y into K segments defined by a set of boundaries $\mathcal{T} = \{\tau_1, \tau_2, \dots, \tau_{K-1}\}$, where $\tau_0 = 0$ and $\tau_K = N$.

To determine the optimal boundary placements for a fixed number of segments K , we employ a kernelized Dynamic Programming algorithm. We utilize a Radial Basis Function (RBF) kernel to map the centroid sequence into a high-dimensional space. The algorithm then finds the globally optimal partition $\hat{\mathcal{T}}_K$ by minimizing the sum of segment

costs $c_{rbf}(\cdot)$:

$$\hat{\mathcal{T}}_K = \arg \min_{\mathcal{T}} \sum_{j=1}^K c_{rbf}(y_{\tau_{j-1}:\tau_j})$$

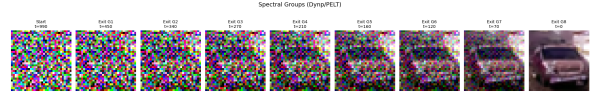


Figure 3. Denoising Trajectory Using Spectral Groups As Sample Points

To then evaluate the empirical fit of the candidate segmentation $\hat{\mathcal{T}}_K$, we map the bounded segments back to the Euclidean space and compute the Residual Sum of Squares (RSS):

$$RSS(\hat{\mathcal{T}}_K) = \sum_{j=1}^K \sum_{i=\tau_{j-1}}^{\tau_j-1} (y_i - \mu_j)^2$$

where μ_j represents the empirical mean of the j -th segment.

Because Dynamic Programming requires a predefined K , we must determine the optimal number of segments without overfitting the data. We achieve this by performing an exhaustive search over $K \in [1, K_{max}]$ and evaluating each segmentation using the Bayesian Information Criterion (BIC). The BIC balances the goodness-of-fit against the complexity of the partition:

$$BIC(K) = N \ln \left(\max \left(\frac{RSS(\hat{\mathcal{T}}_K)}{N}, \epsilon \right) \right) + K \ln(N)$$

Finally, the optimal segmentation is determined by the specific number of segments K^* that yields the minimum BIC:

$$K^* = \arg \min_K BIC(K)$$

The results of this are visualized for a 100 time-step denoising trajectory in Fig.4. Moreover, a comparative denoising trajectory using the endpoints of the groupings is visualized in Fig.3, showing the improved nature of the groupings.

Spectral drift correction. The recurrence above motivates correcting quantization error at the level of the denoising trajectory, rather than only minimizing layer-wise reconstruction error. In diffusion sampling, the denoising network predicts a noise component $\epsilon_\theta(x_t, t)$, which is used by the sampler to compute x_{t-1} . Let $\epsilon_\theta^{FP}(x_t, t)$ denote the full-precision prediction and $\epsilon_\theta^Q(x_t, t)$ denote the quantized prediction. The prediction drift is

$$\Delta_t(x_t) = \epsilon_\theta^{FP}(x_t, t) - \epsilon_\theta^Q(x_t, t).$$

If this error has a systematic timestep-dependent component, then directly compensating for it can reduce accumulated trajectory drift.

To estimate this drift, we use a small calibration set and run both the full-precision and quantized models on the same noisy inputs (x_t, t) . Since our method already groups time-steps according to spectral activation behavior, we estimate a correction separately for each spectral group. Let $G(t)$ denote the spectral group assigned to timestep t .

For compactness, define

$$q_{j,c} = \epsilon_\theta^Q(x_t, t)[c], \quad f_{j,c} = \epsilon_\theta^{FP}(x_t, t)[c], \quad j = G(t),$$

where $q_{j,c}$ and $f_{j,c}$ collect all values for channel c over calibration samples, spatial locations, and time-steps assigned to group j . We fit a channel-wise affine map

$$f_{j,c} \approx a_j[c]q_{j,c} + b_j[c].$$

The corrected quantized prediction is then

$$\hat{\epsilon}_\theta^Q(x_t, t)[c] = a_{G(t)}[c] \epsilon_\theta^Q(x_t, t)[c] + b_{G(t)}[c].$$

This correction is applied only at the denoiser output and therefore does not modify the quantized weights or activations.

For each spectral group j and channel c , the affine parameters are estimated by least squares:

$$a_j[c], b_j[c] = \arg \min_{a,b} E \left[(f_{j,c} - a q_{j,c} - b)^2 \right].$$

This gives the closed-form solution

$$a_j[c] = \frac{Cov(q_{j,c}, f_{j,c})}{Var(q_{j,c}) + \varepsilon}, \quad b_j[c] = E[f_{j,c}] - a_j[c]E[q_{j,c}].$$

Here, all expectations, variances, and covariances are computed over calibration samples, spatial locations, and time-steps in spectral group j .

During sampling, we replace the raw quantized prediction with the corrected prediction:

$$\epsilon_\theta^Q(x_t, t) \longrightarrow \hat{\epsilon}_\theta^Q(x_t, t).$$

For a DDPM-style reverse update, this means replacing $\epsilon_\theta^Q(x_t, t)$ with $\hat{\epsilon}_\theta^Q(x_t, t)$:

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \hat{\epsilon}_\theta^Q(x_t, t) \right) + \sigma_t z.$$

Outlier suppression via timestep-dependent rotations.

In addition to spectral grouping, we also study whether timestep-dependent orthogonal rotations can suppress outliers in either weights or activations before quantization. The motivation is that outliers often dominate the quantization range. If a small number of channels or spatial locations have unusually large magnitude, the quantization scale gets dominated by these magnitudes and possibly a large majority of the distribution all fall or collapse into one or two buckets, increasing rounding error for the majority of the distribution of either weights or activations. Orthogonal rotations preserve the Euclidean norm of vectors.

Let $x \in R^m$ denote an activation along the dimension. A fixed rotation $R_0 \in R^{m \times m}$ can be absorbed into the following weight matrix offline:

$$\widetilde{W} = W R_0^\top.$$

Then, in full precision,

$$\widetilde{W} R_0 x = W R_0^\top R_0 x = W x,$$

so the transformation does not change the model output. In practice, R_0 can be chosen as a structured Hadamard rotation or as a calibrated orthogonal matrix. The purpose of R_0 is to spread the activation or smoothen the distribution before quantization.

However, diffusion activations are timestep-dependent, so a single fixed rotation may not be optimal for every denoising step. We therefore consider a small timestep-dependent correction constructed from Householder reflection matrices. For spectral group $j = G(t)$, we define

$$H_j = H_{j,1} H_{j,2} \cdots H_{j,k},$$

where each Householder reflection is

$$H_{j,\ell} = I - 2 \frac{v_{j,\ell} v_{j,\ell}^\top}{\|v_{j,\ell}\|_2^2 + \varepsilon}.$$

Here $v_{j,\ell} \in R^m$ is a learnable reflection vector for spectral group j and reflection index ℓ . Since each $H_{j,\ell}$ is orthogonal,

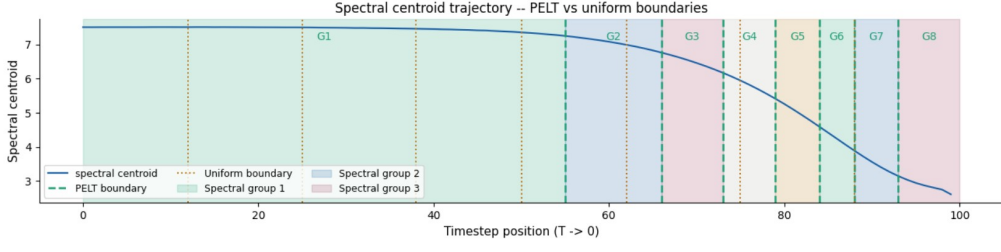


Figure 4. Spectral Groupings

their product H_j is also orthogonal. This guarantees that the rotation preserves activation norm:

$$\|H_j x\|_2 = \|x\|_2.$$

A naive approach would be to apply the timestep-dependent rotation only to the activation while keeping the weight matrix fixed as $\widetilde{W} = WR_0^\top$. This is not function-preserving, because

$$\widetilde{W}R_0H_jx = WH_jx \neq Wx$$

unless $H_j = I$. To avoid changing the full-precision model, we instead use the Householder correction in a way that preserves the output or functional form:

$$\begin{aligned} z &= R_0x, \\ u &= H_{G(t)}z, \\ \hat{u} &= Q(u), \\ \hat{z} &= H_{G(t)}^\top \hat{u}, \end{aligned}$$

and the layer output is computed as

$$y = \widetilde{W}\hat{z}.$$

In the absence of quantization, this reduces exactly to the original rotated layer:

$$H_{G(t)}^\top H_{G(t)} R_0x = R_0x.$$

Thus, the timestep-dependent rotation affects only the coordinate system in which quantization is performed, while preserving the underlying computation.

Householder parameterization and initialization. For each spectral group j , we store k Householder vectors

$$\{v_{j,1}, v_{j,2}, \dots, v_{j,k}\}.$$

The total number of additional parameters is therefore

$$K \cdot k \cdot m,$$

where K is the number of spectral timestep groups. In our implementation, k is kept small, typically $k = 1$ or $k = 2$,

so the parameter overhead remains negligible relative to the U-Net.

The vectors are initialized close to an identity correction. Specifically, each reflection vector is initialized from a small random Gaussian distribution,

$$v_{j,\ell} \sim \mathcal{N}(0, \sigma^2 I),$$

and then normalized before constructing the reflection:

$$\bar{v}_{j,\ell} = \frac{v_{j,\ell}}{\|v_{j,\ell}\|_2 + \varepsilon}.$$

The corresponding reflection is

$$H_{j,\ell} = I - 2\bar{v}_{j,\ell}\bar{v}_{j,\ell}^\top.$$

In implementation, the full matrix $H_{j,\ell}$ does not need to be explicitly formed. For an activation x , a Householder reflection is applied as

$$H_{j,\ell}x = x - 2\bar{v}_{j,\ell}(\bar{v}_{j,\ell}^\top x),$$

The Householder vectors are calibrated after the base quantized model is fixed. Given calibration activations x_t from timestep group j , we optimize the vectors to reduce activation quantization error in the rotated space:

$$\mathcal{L}_{rot} = E_{x_t, G(t)=j} \left[\|Q(H_j R_0 x_t) - H_j R_0 x_t\|_2^2 \right].$$

Only the Householder vectors are updated during this stage; the diffusion model weights and the base rotation R_0 remain fixed.

This rotation concept was evaluated as an ablation. Although it is theoretically well-defined and preserves the computation, our experiments show that it does not improve FID in this setting. This suggests that the remaining degradation in FID score is better explained by weight quantization error since they are at a lower bit width relative to activations which are usually kept in bit width 8 and trajectory-level drift.



Figure 5. Images Generated Through a Variety Of Methods

3. Results

Table 1. Main PTQ results. Our methods are evaluated using 512 generated samples and 512 real CIFAR-10 reference images. Q-Diffusion is included for context, but uses a separate sampling and FID evaluation pipeline, so its absolute FID is not directly comparable. Lower FID is better.

Method	Precision	FID
FP32	W32A32	83.61
Uniform PTQ	W8A8	89.64
Spectral PTQ	W8A8	87.26
Spectral PTQ + Mean Drift	W8A8	88.09
Spectral PTQ + Affine Drift	W8A8	88.75
Spectral PTQ + Householder	W8A8	89.89
Uniform PTQ	W4A8	91.82
Spectral PTQ	W4A8	91.53
Spectral PTQ + Weight Clipping	W4A8	90.03
Spectral PTQ + Weight Clipping + Mixed W4/W8, A8 Mixed Prec.		89.88
Spectral PTQ + Weight Clipping + Mixed W4/W8, A8 Mixed Prec. + Mean Drift		89.04
Spectral PTQ + Weight Clipping + Mixed W4/W8, A8 Mixed Prec. + Affine Drift		90.96
Spectral PTQ + Householder	W4A8	93.33
Q-Diffusion FP32 [†]	W32A32	56.32
Q-Diffusion [†]	W8A8	56.19
Q-Diffusion [†]	W4A8	56.38

[†]Q-Diffusion results are measured using its separate sampling and FID evaluation script. They are included only as a contextual baseline and should not be directly compared by absolute FID to the rows above.

Q-Diffusion is included as a baseline because it is a strong existing PTQ method for diffusion models. However, its reported FID values were obtained using its own generation and evaluation script, which differs from the pipeline used for our main experiments. Therefore, the absolute FID values for Q-Diffusion are not directly comparable to our measured FID values. Instead, we interpret Q-Diffusion mainly by its relative degradation from its own FP32 baseline. Under its pipeline, W8A8 and W4A8 remain very close to FP32.

4. Discussion

Table 1 shows that timestep-aware calibration is important for post-training quantization of diffusion models. In the W8A8 setting, uniform timestep grouping obtains an FID of 89.64, while our spectral grouping strategy improves this to 87.26. This suggests that grouping timesteps according to the frequency structure of the denoising trajectory produces more suitable quantization regions than assigning timesteps uniformly. The improvement supports our main hypothesis: diffusion activations are not stationary across the reverse process, and their spectral evolution across timesteps provides useful information for quantization calibration.

The gains from spectral grouping are smaller in the W4A8 setting. Uniform W4A8 quantization obtains an FID of 91.82, while spectral grouping improves it slightly to 91.53. This indicates that at lower weight precision, the major reason of degradation shifts from activation quantization to weight quantization. In W8A8, both weights and activations retain enough precision that better timestep grouping can noticeably improve the activation quantization behavior. However, when weights are reduced to 4 bits, the quantized weight representation becomes a stronger bottleneck. This also highlights why spectral grouping alone is less effective for W4A8 than for W8A8.

To address the weight-side bottleneck, we evaluated layer-wise weight clipping and mixed precision. Layer-wise clipping improves W4A8 spectral PTQ from 91.53 to 90.03, showing that a single global clipping rule is not optimal for all layers. Different layers have different weight distributions and sensitivities, so calibrating clipping thresholds per layer reduces unnecessary quantization error. Mixed precision further improves the result to 89.88 by keeping the most sensitive layers at higher weight precision. This result suggests that a small number of layers contribute to the final generation degradation, and preserving those layers at 8-bit precision is more effective than applying a uniform 4-bit weight constraint everywhere.

The drift correction experiments provide a trajectory-level view of quantization error. In W4A8, adding mean drift correction to the weight-clipped mixed-precision model gives the best result, reducing FID to 89.04. This supports the idea that quantized diffusion models can suffer from systematic

denoiser-output bias, and that correcting the average prediction drift can improve alignment with the full-precision trajectory. Mean drift correction is simple because it only estimates an additive per-channel correction for each spectral timestep group. In contrast, affine drift correction performs worse, reaching 90.96 in the W4A8 mixed-precision setting. Although affine correction is more expressive because it estimates both scale and bias, it may be more sensitive to calibration noise. With a limited calibration set, the estimated scale factors can overfit the calibration trajectory and introduce additional error during sampling.

The Householder rotation ablation did not improve performance. In W8A8, spectral PTQ with Householder rotations obtains an FID of 89.89, which is worse than spectral PTQ alone. In W4A8, the Householder variant reaches 93.33, also worse than the corresponding spectral baseline. This negative result suggests that the remaining error in our setting is not primarily caused by an activation outlier. Although orthogonal rotations are theoretically attractive because they preserve norm while redistributing activation distributions to more smoother distributions, our results indicate that applying timestep-dependent rotations around the activations does not reduce the errors that matter most for final sample quality.

The Q-Diffusion results are included only as a baseline. Under its own sampling and evaluation pipeline, Q-Diffusion preserves FID very close to its FP32 baseline for both W8A8 and W4A8. However, these absolute FID values are not directly comparable to our results because they are produced using a separate generation and FID evaluation setup. Therefore, we interpret Q-Diffusion mainly by its relative degradation from its own FP32 baseline rather than by comparing its raw FID values directly to ours.

Overall, the results indicate that spectral grouping is most useful when activation quantization remains a significant source of error, as in W8A8. For more aggressive W4A8 quantization, weight-side calibration and selective mixed precision become more important. Drift correction can further improve the best W4A8 setting when implemented as a stable mean correction, but more seemingly reasonable corrections such as affine mappings or timestep-dependent rotations may require larger calibration sets. These findings suggest that effective diffusion PTQ should combine timestep-aware activation calibration with targeted weight-side techniques and trajectory-level correction.

5. Conclusion

This work investigated spectral grouping as a principled approach to post-training quantization of diffusion models, alongside two complementary mechanisms: drift correction and timestep-dependent Householder rotations for outlier

suppression. Our central finding is that grouping timesteps according to the frequency-domain evolution of the denoising trajectory produces more suitable quantization regions than uniform timestep assignment, yielding a meaningful FID improvement in the W8A8 setting ($89.64 \rightarrow 87.26$). This confirms that the spectral non-stationarity of diffusion activations is a practically useful signal for quantization calibration, not merely a theoretical curiosity.

Several important limitations temper these conclusions. First, all experiments were conducted on CIFAR-10 with a relatively small U-Net, and it is unclear whether spectral grouping scales favorably to larger architectures such as latent diffusion models operating on higher-resolution data, where activation distributions and spectral dynamics may differ substantially. Second, our FID evaluations use only 512 generated and 512 reference samples, which introduces considerable variance into our estimates. More reliable conclusions would require evaluating over several thousand samples, though this was constrained by available compute. Third, the calibration set used for both spectral grouping and drift correction is small, which likely explains why affine drift correction underperforms mean drift correction despite being more expressive — the scale estimates overfit the calibration trajectory and introduce error during sampling. Larger or more diverse calibration sets may be necessary for more expressive corrections to realize their potential.

The negative result for Householder rotations is informative. Although timestep-dependent orthogonal rotations are theoretically well-motivated and norm-preserving, they did not improve FID under any configuration tested. This suggests that in the W4A8 regime, the dominant source of degradation is weight quantization error rather than activation outliers, and that redistributing activation energy across channels does not address this bottleneck. Future work targeting outlier suppression in diffusion models should therefore focus on the weight side, potentially through learned per-channel smoothing factors that vary with timestep, or through weight rounding methods that explicitly account for the downstream trajectory impact of individual weight errors.

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